



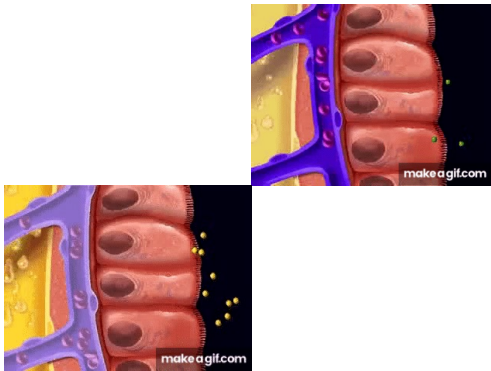
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HAWLER MEDICAL UNIVERSITY

Physiology of Small Intestine

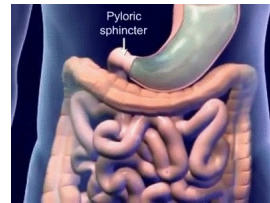
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Contents:

- Introduction to small intestine.
- Small intestine regions and their functions.
- Types of movement within the small intestine.
- Villi, Microvilli, and Circular folds.
- Brush border enzymes.
- Intestinal digestion and absorption of:
 - Carbohydrates, Proteins, and Lipids.
 - Electrolytes, Vitamins, and Water.



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Objectives:

After studying this lecture, you will be able to:

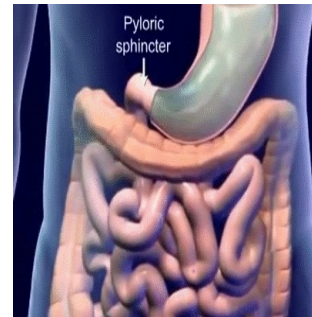
- Describe the small intestine.
- List the regions of the small intestine and their functions.
- Describe the importance of villi, microvilli, & circular folds.
- List the Brush border enzymes.
- Describe the intestinal digestion and absorption of nutrient molecules.



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Introduction: The small intestine:

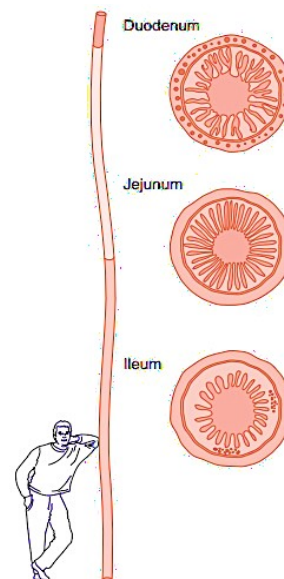
- The small intestine is the **longest part** of the digestive system.
- It extends from the **pyloric sphincter** to the **ileocecal junction**.



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Introduction: The small intestine:

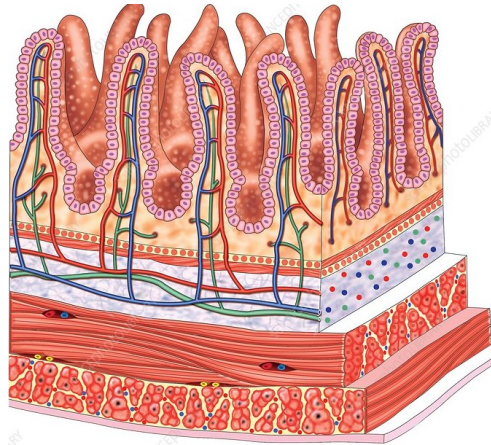
- Its length is:
 - ~3-4 m. long (**living person**)
 - ~6.5 m. long (**cadaver**)
- Average diameter= 2.5 cm.
- Small intestine is the principal site for **nutrient digestion** and **absorption**.



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Introduction: The small intestine:

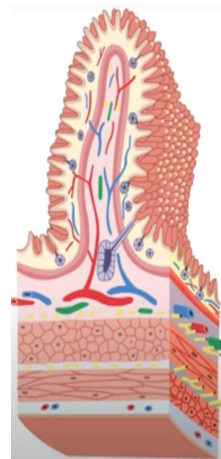
- The wall of the small intestine has (4) layers:
 - **Mucosa,**
 - **Submucosa,**
 - **Muscularis,**
 - **Serosa.**



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Introduction: The small intestine:

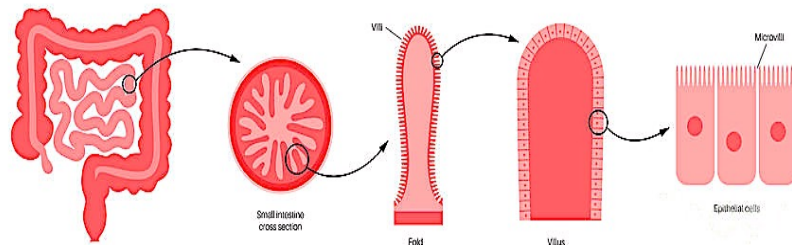
- The **mucosa** contains **villi** and **microvilli (increasing surface area)**
- The **submucosa** houses **Brunner's (duodenal) glands (alkaline mucus)**
- The **muscularis** consists of **circular and longitudinal smooth muscle, essential for motility.**



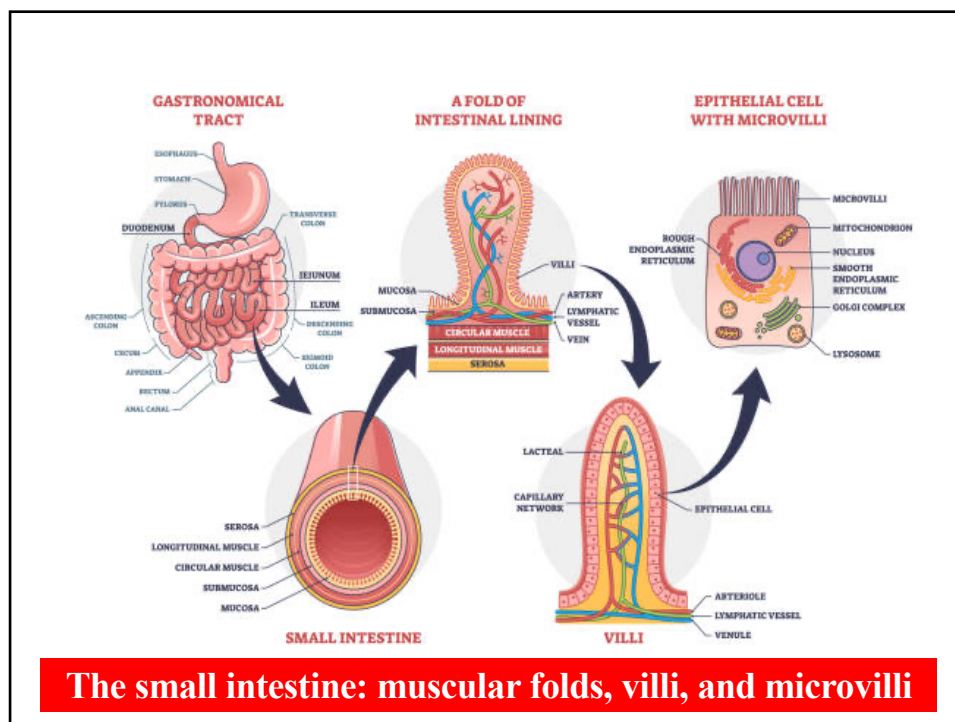
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Introduction: The small intestine:

- It is **functionally adapted** for **efficient nutrient breakdown, absorption, and motility**.
- It includes **folds, villi, and microvilli**, which together form an **extensive absorptive surface area**.
- **Capillaries and lacteals** within villi **transport absorbed nutrients into circulation**.



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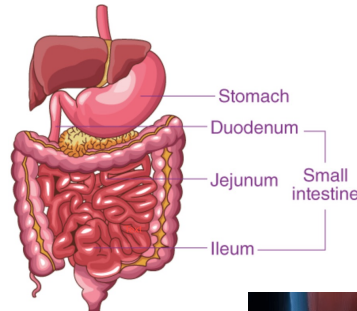
The small intestine: muscular folds, villi, and microvilli

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Regions of the Small Intestine:

The small intestine is divided into 3 regions:

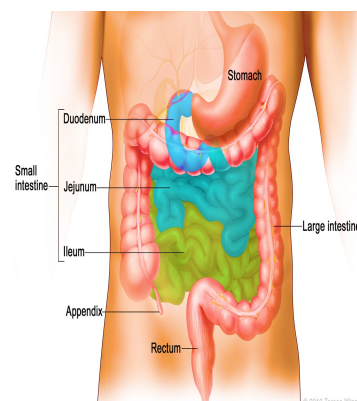
- **Duodenum:**
- **Jejunum:**
- **Ileum:**



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Duodenum:

- The first and the **shortest region** (~25 cm long).
- It curves around the head of the pancreas.
- Plays a central role in receiving chyme and digestive secretions.
- It receives secretions from the **liver**, **gallbladder**, and **pancreas**, which aid digestion.



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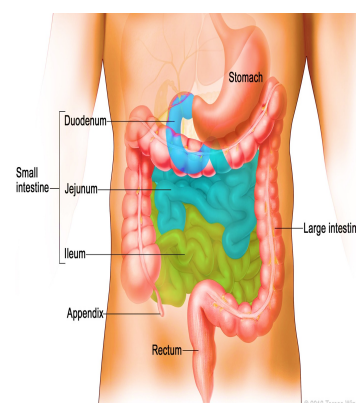
Duodenum:

- **Brunner's (Duodenal) glands:**
 - They are located in the duodenal submucosa and secrete **alkaline mucus**, that:
 - **neutralize acidic chyme** entering from the stomach.
 - **provide an optimal pH** for pancreatic enzyme activity.
 - **prevents damage** to the intestinal mucosa.
- Duodenum also contains endocrine cells that secrete **Secretin** and **Cholecystikin**.

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Jejunum:

- It is located in the middle and is especially **specialized for absorption**.
- It accounts for **~2/5th** of the small intestine.
- It is **richly supplied with blood vessels** and has **prominent circular folds (plicae circulares)**.



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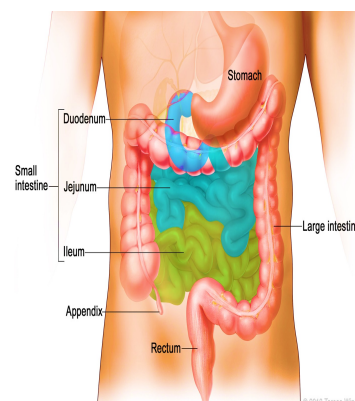
Jejunum:

- Its mucosa contains abundant **villi** that **maximize nutrient absorption**.
- The jejunum is the main site for **absorption** of:
 - **Carbohydrates,**
 - **Proteins,**
 - **Lipids.**

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Ileum:

- It is the **longest portion** ~ $\frac{3}{5}$ th of the small intestine.
- Ileum has fewer folds and villi compared to the jejunum, but it still **contributes significantly to absorption**.



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Ileum:

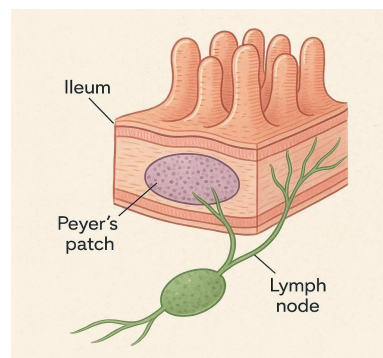
- It is specialized for the uptake of **bile salts** and **vitamin B12**, which are essential for **digestion** and **hematopoiesis**.
- The ileum ends at the **ileocecal valve**, which **regulates the passage of contents into the large intestine**.



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Ileum: Peyer's patches:

- **Peyer's patches** are lymphoid aggregates found in the ileum.
- They provide **immune surveillance** against pathogens entering with food.
- They contain **B** and **T** lymphocytes, **supporting gut-associated immunity**.



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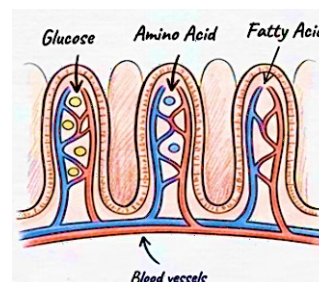
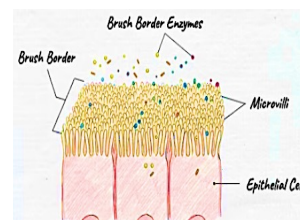
Intestinal Juice:

- a watery mixture containing **mucus** and **enzymes** secreted by the glands of the small intestine.
- It **provides the medium in which digestive processes occur.**
- The **enzymes** complete the digestion of **carbohydrates and proteins.**
- The **secretions (mucus)** lubricate the intestinal walls and facilitate transport of nutrients.

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Brush Border Enzymes:

- The brush border contains enzymes such as **lactase**, **maltase**, **sucrase**, and **peptidases**.
- They are bound to the **microvilli membranes** rather than secreted into the lumen.
- **They allow final hydrolysis of:**
 - Carbohydrates → Glucose
 - Peptides → amino acids.
 - Lipids → Fatty acids



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Motility Patterns of the Small Intestine:

- The small intestine exhibits **segmentation** and **peristalsis**.

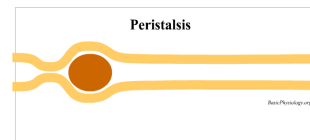
- **Segmentation:**

- involves local contractions that **mix chyme with digestive secretions**.



- **Peristalsis:**

- involves waves of contraction that **move chyme forward and then maximizes digestion & absorption**.



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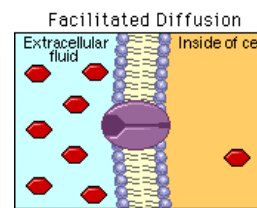
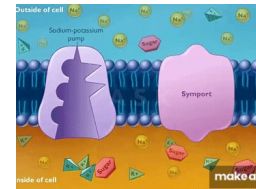
Digestion of Carbohydrates:

- Carbohydrate digestion begins with **salivary** and **pancreatic amylase**.
- **Amylase** breaks down **polysaccharides** into **disaccharides**.
- Brush border enzymes convert **disaccharides** into **monosaccharides**.
- Only **monosaccharides** like **glucose**, **galactose**, and **fructose** are **absorbed**.

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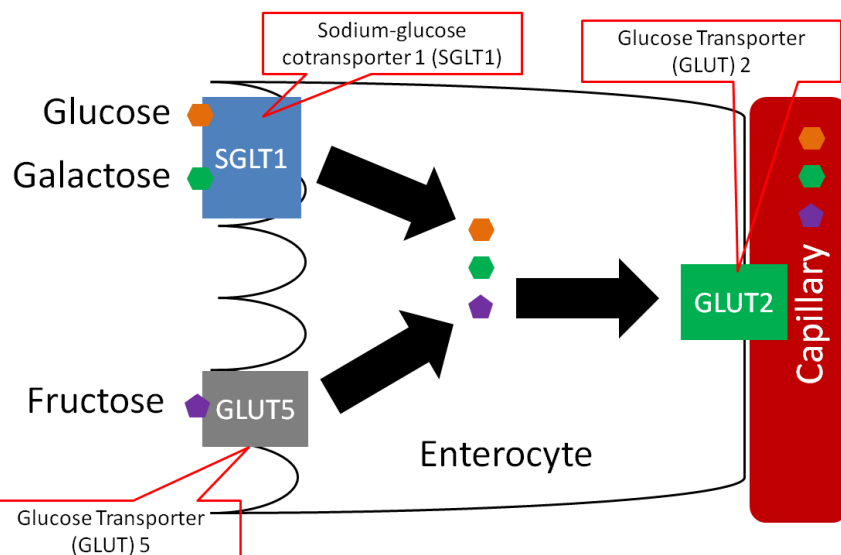
Absorption of Carbohydrates:

- Glucose and galactose are absorbed via **sodium-dependent co-transport**.
- Fructose is absorbed by **facilitated diffusion**.
- Eventually, all monosaccharides enter the bloodstream.



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Absorption of Carbohydrates:



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Digestion of Proteins:

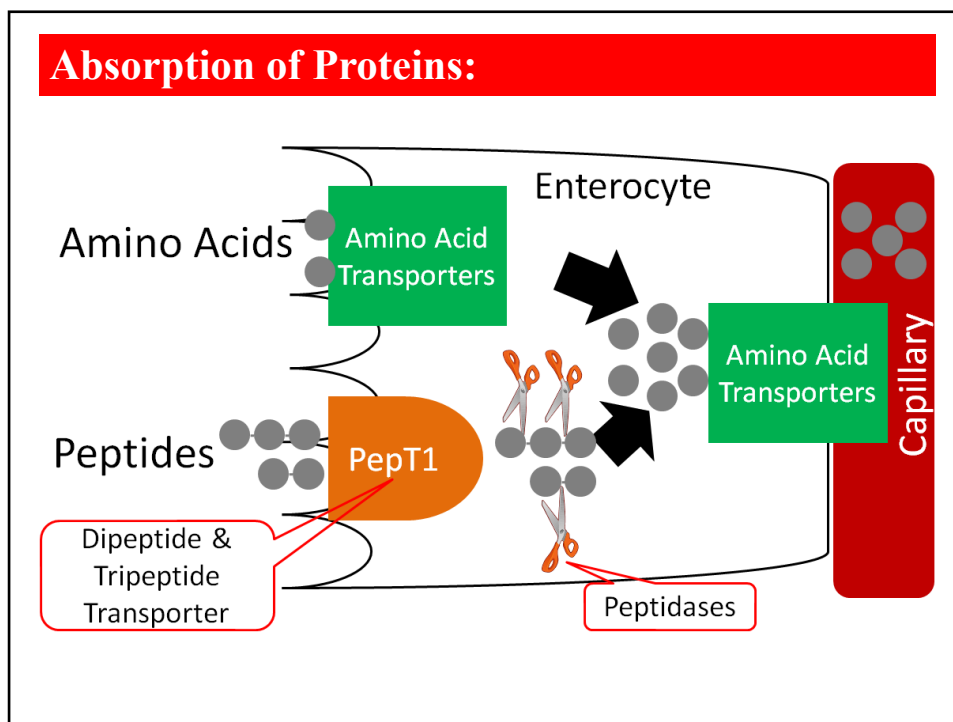
- Proteins are initially digested in the stomach by **pepsin**
- In the small intestine, pancreatic enzymes (**trypsin**, **chymotrypsin**, and **carboxypeptidase**) further **hydrolyze** proteins.
- Brush border **peptidases** complete the breakdown of proteins to **amino acids (AAs)**.
- Both **AAs** and **small peptides** can then be **absorbed**.

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Absorption of Proteins:

- **AAs** are absorbed by **Na⁺ dependent active transport systems**.
- **Dipeptides** and **tripeptides** are absorbed via proton-coupled peptide transporters (**PepT1**).
- Once inside the **enterocyte**, dipeptides and tripeptides are hydrolyzed into **AAs**.
- Absorbed **AAs** enter the blood and are transported to the liver.

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Digestion of Lipids:

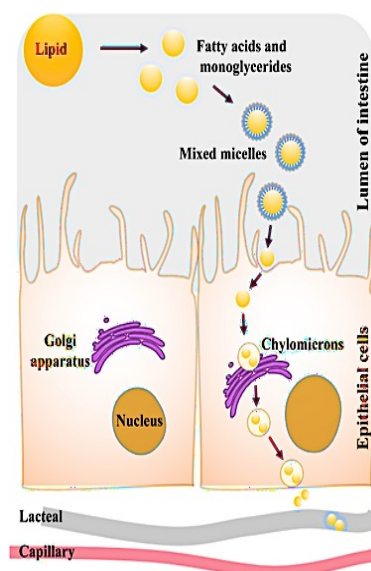
- Lipids require **emulsification** by bile salts into **micelles**.
- **Pancreatic lipase** hydrolyzes **triglycerides** into **monoglycerides** and **free fatty acids**.
- **Phospholipase** and **cholesterol esterase** further process dietary fats.
- **Micelles** transport lipid products to the epithelial surface for absorption.

The diagram illustrates the process of emulsification. A large 'Fat Globule' (represented by a cluster of blue spheres) is broken down by 'Bile Salt' (represented by a yellow and green molecule) into smaller 'Emulsified Fat Droplets'. The bile salt molecule is shown with a 'Hydrophobic' end (yellow) and a 'Hydrophilic' end (green). The emulsified fat droplets are shown as smaller clusters of blue spheres, each surrounded by bile salt molecules.

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Absorption of Lipids:

- Lipid products **diffuse** across the intestinal epithelium into **enterocytes**.
- Inside cells, they are re-esterified into triglycerides.
- **Triglycerides** combine with **cholesterol**, **phospholipids**, and **proteins** to form **chylomicrons**.
- **Chylomicrons** are secreted into **lacteals** and transported through the lymphatic system.



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Absorption of Vitamins:

- Fat-soluble vitamins (A, D, E, K) are absorbed with dietary lipids via micelles.
- Water-soluble vitamins use **diffusion** or **carrier-mediated mechanisms**.
- Vitamin B12 requires intrinsic factor for absorption in the **ileum**.
- This highlights the importance of intact intestinal mucosa for micronutrient uptake.

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Absorption of Minerals:

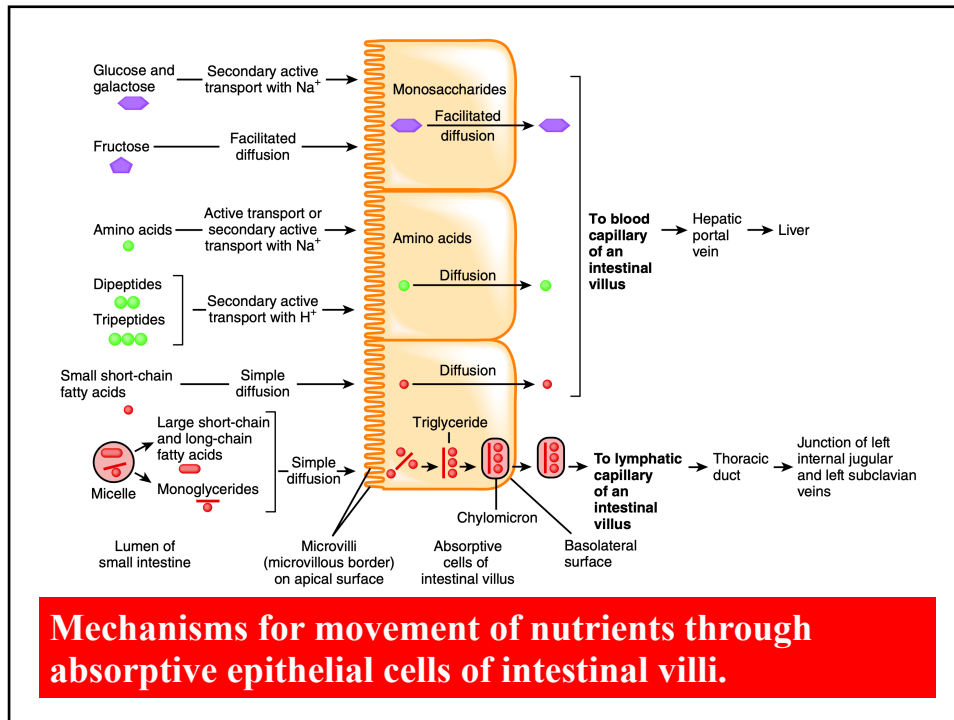
- **Calcium** (Ca^+) **absorption** occurs in the duodenum and **is vitamin D-dependent**.
- **Iron absorption** is **enhanced by gastric acid** and occurs mainly in the duodenum.
- **Magnesium, phosphate, and zinc** are absorbed **throughout the small intestine**.

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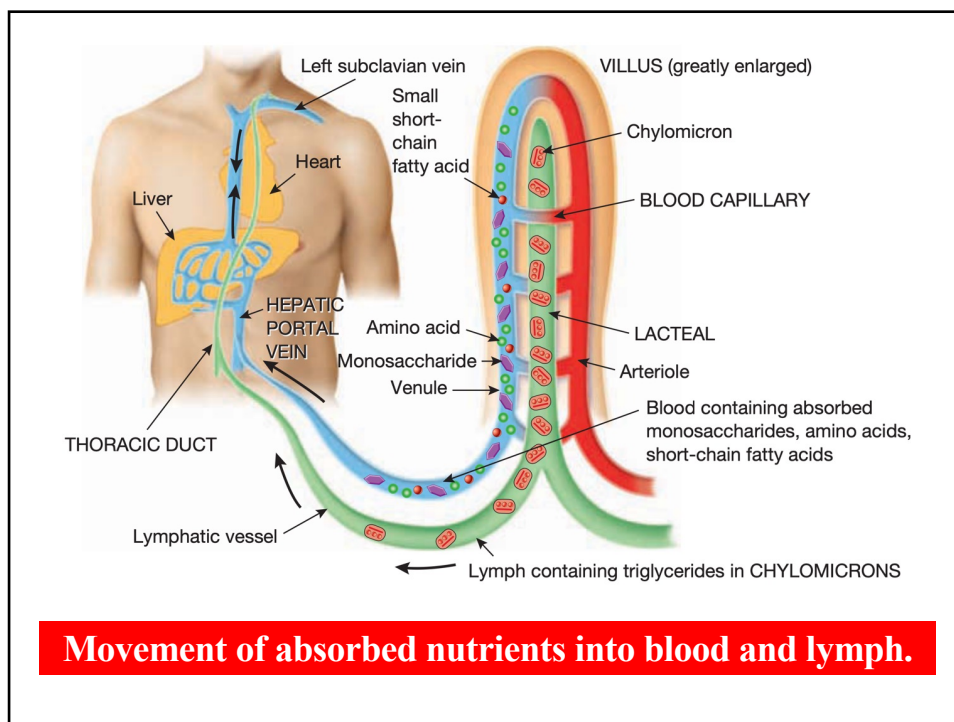
Water and Electrolytes:

- The small intestine absorbs **large volumes of water** and **electrolytes** daily.
- Na^+ absorption is **coupled with nutrient absorption** and also occurs via **Na^+ channels**.
- Cl^- is absorbed **passively** or in **exchange** with HCO_3
- K^+ absorption occurs mainly by **diffusion**.

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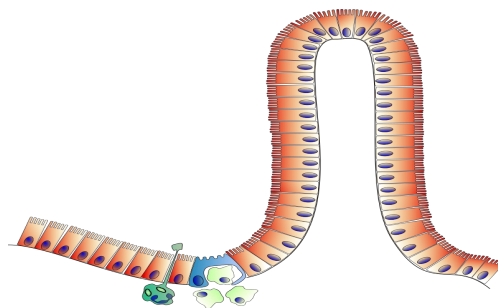
Neural Control:

- The **enteric nervous system** regulates intestinal **motility** and **secretion**.
- **Parasympathetic stimulation** ↑↑ **peristalsis & secretion**
- **Sympathetic stimulation** ↓↓ **motility & blood flow**
- The **coordination** ensures optimal digestion and absorption under varying conditions.

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Immune Function:

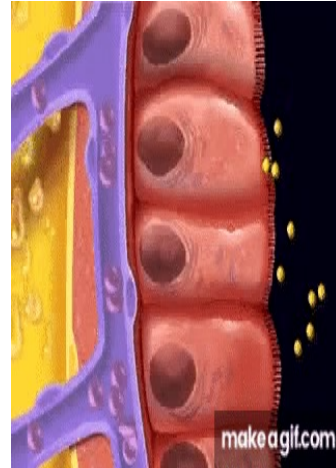
- The small intestine is not only a digestive organ but also an immune barrier.
 - **Peyer's patches** and **intraepithelial lymphocytes** protect against pathogens.



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Functions of the Small Intestine:

1. **Segmentations** mix chyme with digestive juices and bring food into contact with mucosa for absorption; peristalsis propels chyme through small intestine.
2. **Completes digestion** of carbohydrates, proteins, and lipids; begins and completes digestion of nucleic acids.
3. **Absorbs ~90%** of nutrients and water that pass through digestive system.



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Questions/Comments



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